Supplementary Protocol Note

for "Symmetry-Aware Multi-Granularity Feature Balancing for Old Film Restoration"

Purpose of this note

This short note is included to make the submission package self-contained. It does not add new experimental claims beyond those reported in the manuscript. Instead, it clarifies four protocol choices that are referenced in the main text and that are likely to matter during peer review: the synthetic-benchmark significance test, the baseline-adaptation policy, the interpretation of the real-world no-reference metrics, and the complexity-measurement protocol.

1. Synthetic-benchmark statistical test

The manuscript reports paired, two-sided Wilcoxon signed-rank tests on the *per-clip mean* metrics of the 30 clips in the SRWOV synthetic test set. The comparison is performed between MgfrOFR and the strongest baseline in the unified reproduction protocol, MambaOFR. The reported p-values are those stated in the manuscript:

- PSNR: $p = 3.7 \times 10^{-9}$;
- SSIM: $p = 3.5 \times 10^{-3}$;
- LPIPS: $p = 0.71$;
- DISTS: $p = 0.11$.

The purpose of this test is to verify that the observed fidelity improvement is not driven by a small number of outlier clips. No additional significance claim is made for the 216-clip real-world no-reference metrics.

2. Baseline adaptation policy

The manuscript compares MgfrOFR with three groups of baselines: dedicated OFR methods, generic video restoration methods, and enhancement-oriented methods. To improve comparability:

- all methods are evaluated in the same metric environment;
- dedicated OFR methods are reproduced under the same REDS-based synthetic film degradation protocol;
- for generic video restoration methods, the published architecture and default optimization recipe are preserved, while the task-specific degradation generation is replaced so that the methods can be trained on OFR-style inputs;
- enhancement-oriented methods are kept as reference systems for perceptual trade-offs and should not be interpreted as perfectly task-matched OFR baselines.

The comparison is therefore intended as a unified reproduction study under a common degradation protocol, not as a claim that every baseline has been exhaustively re-optimized for the OFR task.

3. Real-world no-reference metrics

The real-world SRWOV evaluation reports NIQE, BRISQUE, and CLIP-IQA+ on 216 archival clips without ground truth. These metrics are included because they emphasize different attributes of restoration quality: NIQE and BRISQUE quantify statistical naturalness, whereas CLIP-IQA+ is more sensitive to perceptual sharpness and high-level visual appeal. For this reason, the manuscript reports the three metrics separately and interprets them descriptively rather than collapsing them into a single superiority claim.

4. Complexity-measurement protocol

The complexity table in the manuscript mixes directly measured values and published reference values. The protocol is as follows:

- MgfrOFR, MambaOFR, and RTN are measured with PyTorch FlopCounterMode on a $1 \times 7 \times 3 \times 256 \times 256$ input, including the full seven-frame forward pass and the RAFT optical-flow front end;
- RRTN and DeepRemaster use published Params/FLOPs where direct measurement is not compatible with the unified measurement path;
- training memory and speed are profiled on NVIDIA L40 GPUs with identical batch size, and speed is reported as the wall time of one mock training step divided by the seven frames in the clip.

This protocol is intended to make the overhead of MgfrOFR interpretable relative to the baseline backbone while keeping the reported numbers transparent.

5. Materials included in the package

The submission package contains the revised manuscript PDF, the source LaTeX files, this protocol note, and a revised cover letter. Code, configurations, and evaluation scripts are publicly available at `https://github.com/wscffaa/MgfrOFR`.